

Large batch distributed training. CPU offloadings. Quantization. Mixed Precision Training

Seminar

Optimization for ML. Faculty of Computer Science. HSE University

Accurate, Large Minibatch SGD. Motivation

💡 Main Pros of Big Data

The increasing data and model scale is rapidly improving accuracy

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Solution: Use distributed SGD with large batch size to make **more efficient** iterations

Accurate, Large Minibatch SGD. Problem

Loss function

$$L(w) = \frac{1}{|X|} \sum_{x \in X} l(x, w)$$

One Iteration of Minibatch SGD (batch size is n)

$$w_{t+1} = w_t - \eta \frac{1}{n} \sum_{x \in \mathcal{B}} \nabla l(x, w_t)$$

k Iterations of Minibatch SGD (batch size is n)

$$w_{t+k} = w_t - \eta \frac{1}{n} \sum_{j < k} \sum_{x \in \mathcal{B}_j} \nabla l(x, w_{t+j})$$

One Large Batch Iteration of Minibatch SGD (batch size is kn)

$$\hat{w}_{t+1} = w_t - \hat{\eta} \frac{1}{kn} \sum_{j < k} \sum_{x \in \mathcal{B}_j} \nabla l(x, w_t)$$

Desired due to multi-GPU training: $\hat{w}_{t+1} \sim w_{t+k}$

Accurate, Large Minibatch SGD. Main idea

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💡 Main Paper Assumption

If we could assume $\nabla l(x, w_t) \sim \nabla l(x, w_{t+j})$ for $j < k$, then setting $\hat{\eta} = k\eta$ would yield $\hat{w}_{t+1} \sim w_{t+k}$

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i Question

When is condition $\nabla l(x, w_t) \sim \nabla l(x, w_{t+j})$ clearly not hold?

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1. The network changes rapidly in initial training
2. Very large k causes very large $\hat{\eta}$ and makes training too unstable

Accurate, Large Minibatch SGD. Solving assumption problems

Question

How would you struggle with assumption problems?

Accurate, Large Minibatch SGD. Solving assumption problems

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How would you struggle with assumption problems?

Gradual warmup. Iteration-wise linear scheduler for start value $\hat{\eta} = \eta$ and finish value $\hat{\eta} = k\eta$ after ~ 5 epochs.

- avoids a sudden increase of the learning rate

Constant per-worker sample size. For global batch size kn we keep the *per-worker* sample size n constant when changing the number of workers k .

Accurate, Large Minibatch SGD. Solving assumption problems

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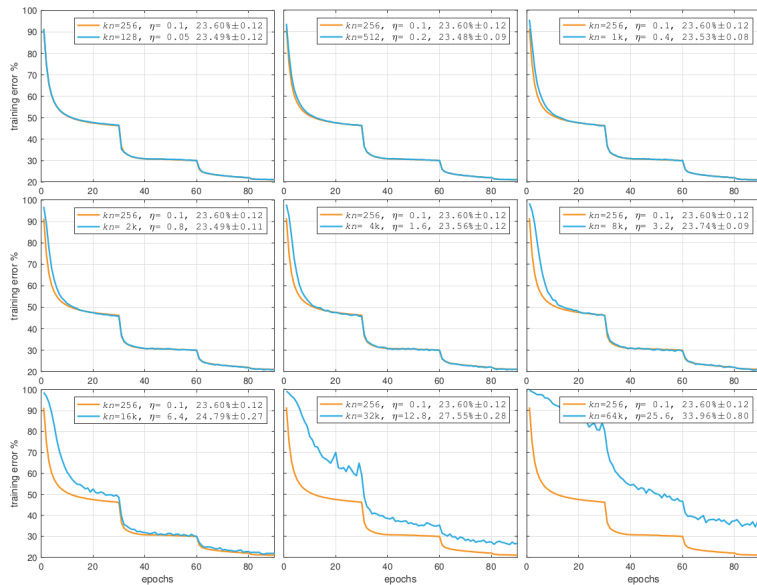
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Constant per-worker sample size. For global batch size kn we keep the *per-worker* sample size n constant when changing the number of workers k .

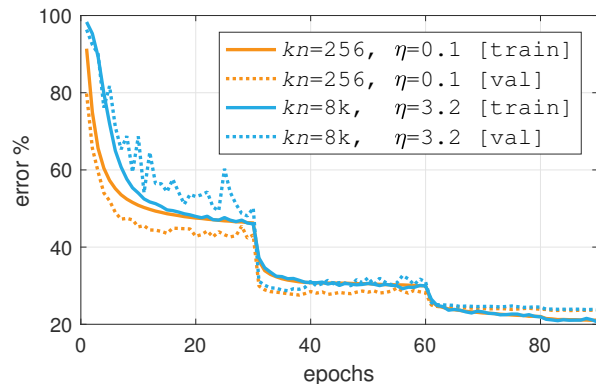
- extremely important for Batch Normalization!

Accurate, Large Minibatch SGD. Results on ImageNet

The training curves closely match the baseline (aside from the warmup period) up through $8k$ minibatches.



Accurate, Large Minibatch SGD. Results on ImageNet



Both sets of curves match closely after training for sufficient epochs.

Note that the BN statistics (for inference only) are computed using running average, which is updated less frequently with a large minibatch and thus is noisier in early training (this explains the larger variation of the validation error in early epochs).

Reduce memory usage. CPU Offloading

- Offloading the weights to the CPU and only loading them on the GPU when performing the forward pass
- CPU offloading works on submodules rather than whole models.
- Inference is much slower due to the iterative uploading and offloading.
- Colab Example  Open in Colab.

Reduce memory usage. Model Offloading

- CPU Offloading makes inference slower because *submodules* are moved to GPU as needed, and they're immediately returned to the CPU when a new module runs.
- Full-model offloading is an alternative that moves whole models to the GPU, instead of handling each model's constituent *submodules*.
- During model offloading, only one of the main components of the pipeline (typically the text encoder, UNet or VAE) is placed on the GPU while the others wait on the CPU.
- Colab Example  Open in Colab.

Reduce memory usage. Quantization

- Quantization maps a floating point value $x \in [\alpha, \beta]$ to a b -bit integer $x_q \in [\alpha_q, \beta_q]$.
- The quantization process is defined as

$$x_q = \text{clip}\left(\text{round}\left(\frac{1}{s}x + z\right), \alpha_q, \beta_q\right)$$

And the de-quantization process is defined as

$$x = s(x_q - z)$$

The value of scale s and zero point z are

$$s = \frac{\beta - \alpha}{\beta_q - \alpha_q} \tag{1}$$

$$z = \text{round}\left(\frac{\beta\alpha_q - \alpha\beta_q}{\beta - \alpha}\right) \tag{2}$$

(3)

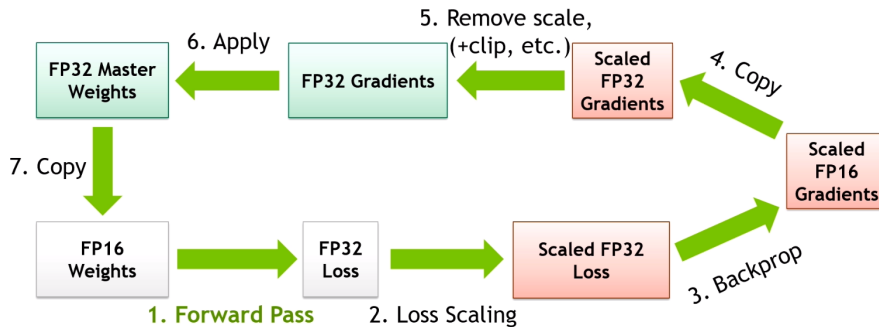
Note that z is an integer and s is a positive floating point number.

- Quantization allows to perform a lot of heavy DL-operations (e.g. matrix multiplication) in integer scope using efficient integer hardware (NVIDIA Tensor Core or Tensor Core IMMA operations) and algorithms.

Reduce memory usage. Quantization

- For more theory look at *Quantization for Neural Networks* , Lei Mao: [git](#).
- Colab Example  Open in Colab.

Mixed Precision Training (MPT)



Mixed Precision Training is a technique where both 16-bit (FP16) and 32-bit (FP32) floating-point types are used during neural network training. This approach accelerates training and reduces memory usage without compromising model accuracy.

Why is MPT beneficial?

- **Faster Training:** Utilizing FP16 allows for faster computations, especially on modern GPUs equipped with Tensor Cores, such as NVIDIA Volta, Turing, and Ampere architectures. This can lead to training speedups of 2–3 times.
- **Reduced Memory Consumption:** FP16 uses half the memory compared to FP32, enabling the training of larger models or the use of larger batch sizes.
- **Maintained Accuracy:** With proper implementation, mixed precision training maintains the same level of model accuracy as full precision (FP32) training.

How does it work?

1. **Autocasting:** Certain operations, like matrix multiplications and convolutions, are performed in FP16, while others that require higher precision remain in FP32. This is managed automatically in frameworks like PyTorch and TensorFlow.
2. **Loss Scaling:** To prevent small gradient values from underflowing in FP16, the loss is scaled up during backpropagation and then scaled down before the optimizer step.

PyTorch *example*:

```
from torch.cuda.amp import autocast, GradScaler

scaler = GradScaler()

for data, target in dataloader:
    optimizer.zero_grad()
    with autocast():
        output = model(data)
        loss = loss_fn(output, target)
    scaler.scale(loss).backward()
    scaler.step(optimizer)
    scaler.update()
```

When to use Mixed Precision Training?

- Training on Modern GPUs: Particularly effective on GPUs with Tensor Cores (e.g., NVIDIA V100, A100).
- Resource-Constrained Environments: Reduces memory consumption and accelerates training, beneficial when computational resources are limited.
- See also an interesting blog post on the topic of Mixed Precision Training 